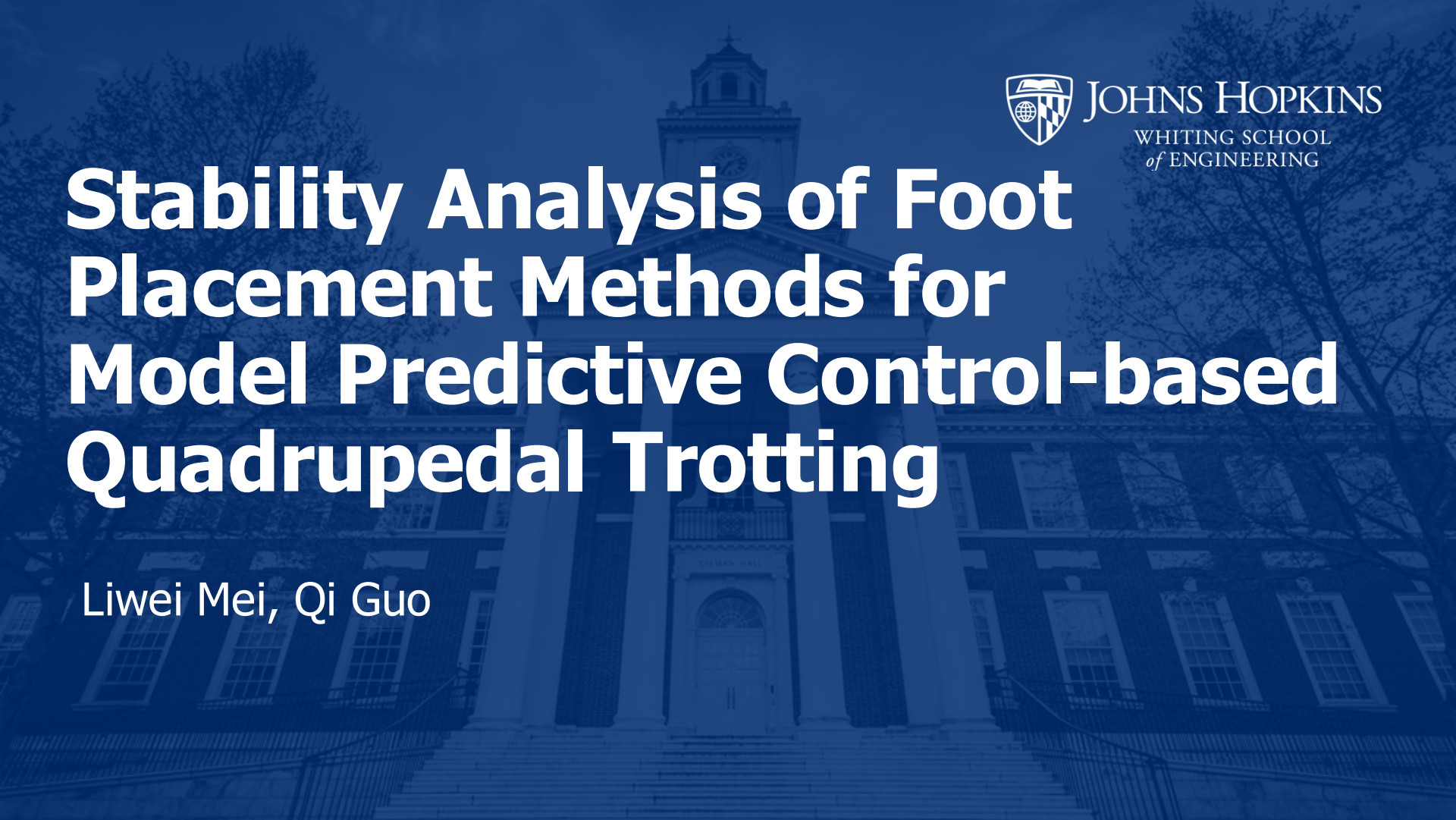


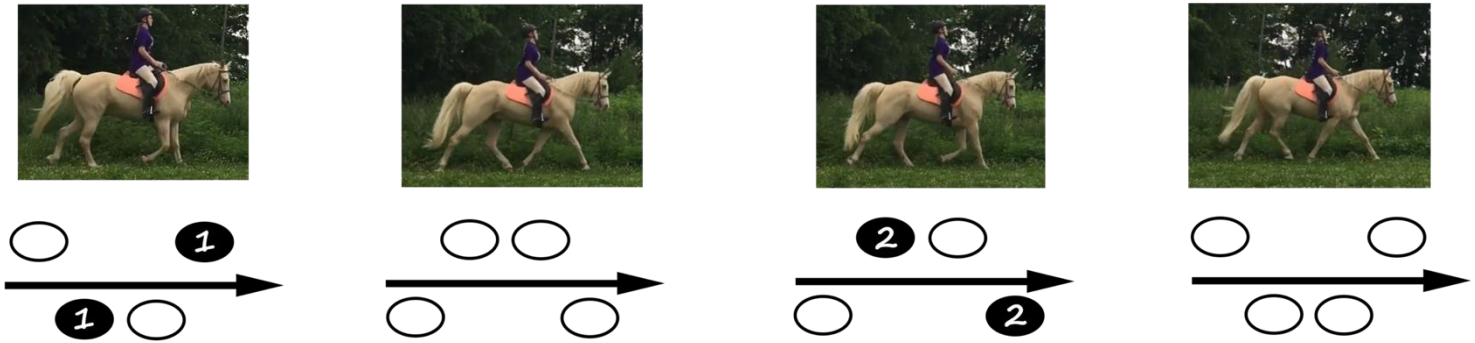
JOHNS HOPKINS
WHITING SCHOOL
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Stability Analysis of Foot Placement Methods for Model Predictive Control-based Quadrupedal Trotting

Liwei Mei, Qi Guo

Research Question:
How Does Swing-Leg Foot Placement Affect Trotting Stability?

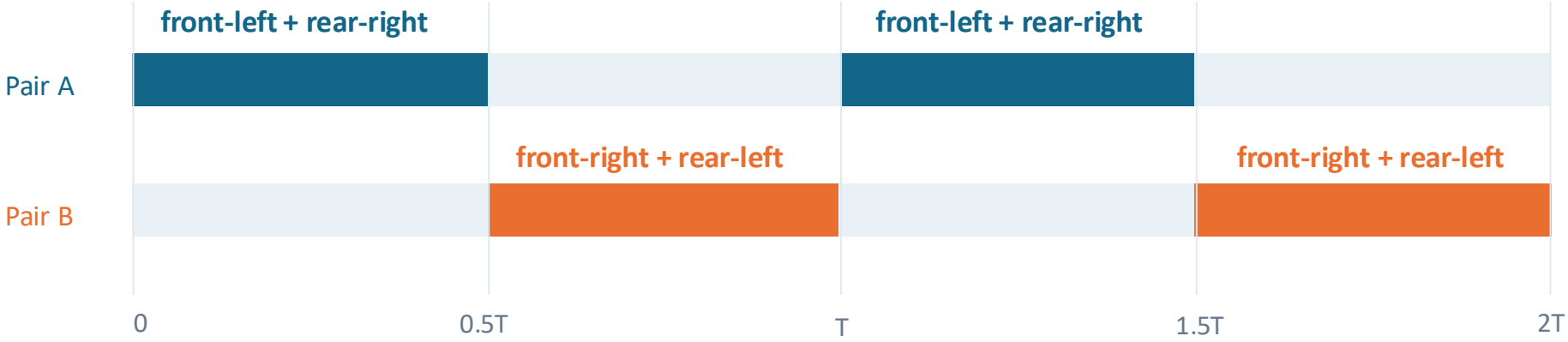
Trotting (alternating diagonal-leg contacts) [4]

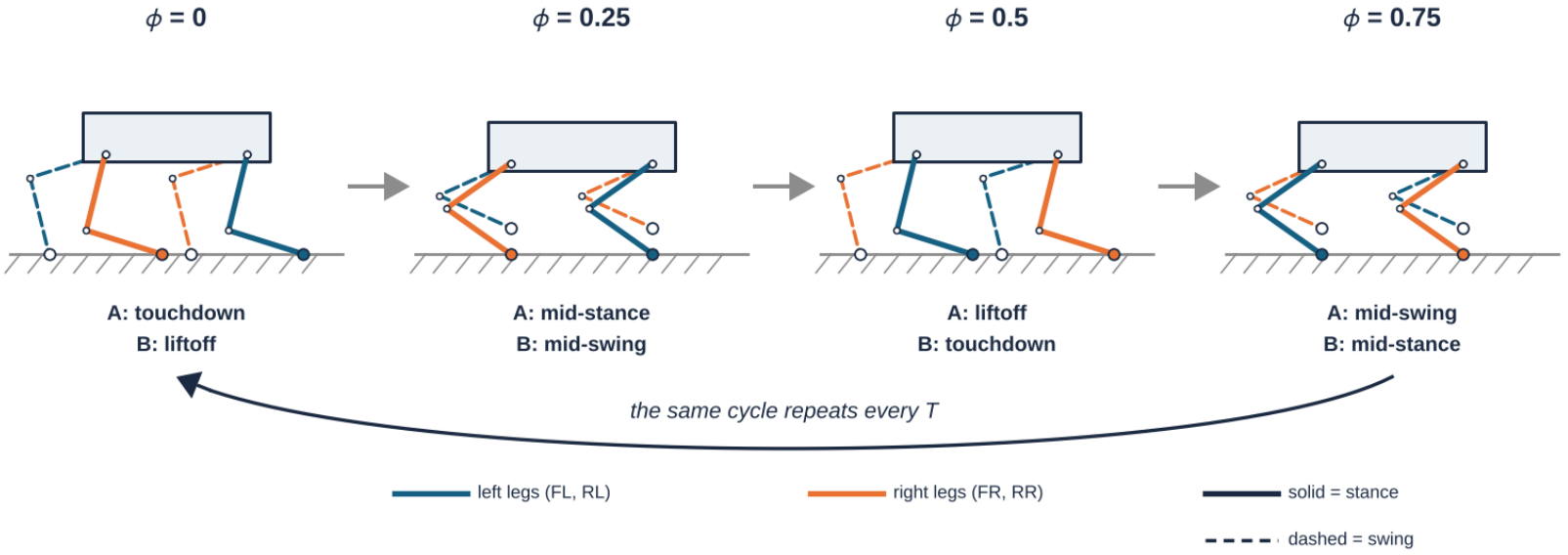


Our focus is how selecting future touchdown locations for swing legs influences stride-to-stride stability.

Phase in Trot

The figure uses an idealized 50% duty cycle and omits the double-support and free-flight phases.




ϕ = 0
ϕ = 0.25
ϕ = 0.5
ϕ = 0.75
A: touchdown
B: liftoff
A: mid-stance
B: mid-swing
A: liftoff
B: touchdown
A: mid-swing
B: mid-stance
the same cycle repeats every T
left legs (FL, RL)
right legs (FR, RR)
solid = stance
dashed = swing

[4]





Model Predictive Control (MPC) [1]

$$m\ddot{\boldsymbol{p}} = \sum_i \boldsymbol{f}_i - m\boldsymbol{g}, \qquad \frac{d}{dt}(\boldsymbol{I}\boldsymbol{\omega}) = \sum_i (\boldsymbol{r}_i - \boldsymbol{p}) \times \boldsymbol{f}_i$$

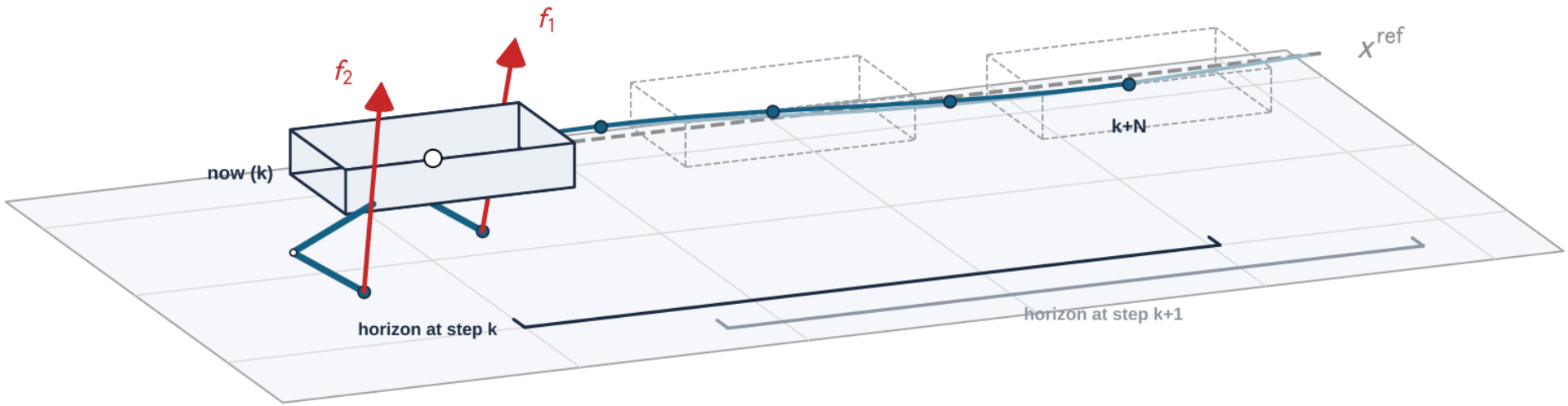
$$\boldsymbol{x}_{j+1} = A_j\,\boldsymbol{x}_j + B_j\,\boldsymbol{f}_j, \qquad j = 0, \ldots, N-1$$

$$\min_{\boldsymbol{f}_0, \ldots, \boldsymbol{f}_{N-1}} \sum_{j=1}^{N} \left\| \boldsymbol{x}_j - \boldsymbol{x}_j^{\text{ref}} \right\|_Q^2 + \sum_{j=0}^{N-1} \left\| \boldsymbol{f}_j \right\|_R^2$$

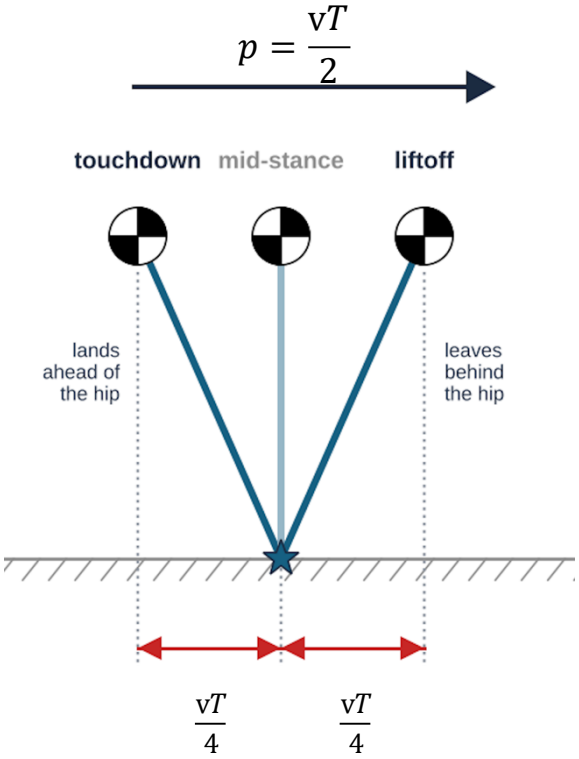
$$\text{s.t.} \quad \boldsymbol{x}_{j+1} = A_j \boldsymbol{x}_j + B_j \boldsymbol{f}_j, \qquad \boldsymbol{x}_0 = \text{measured state}$$

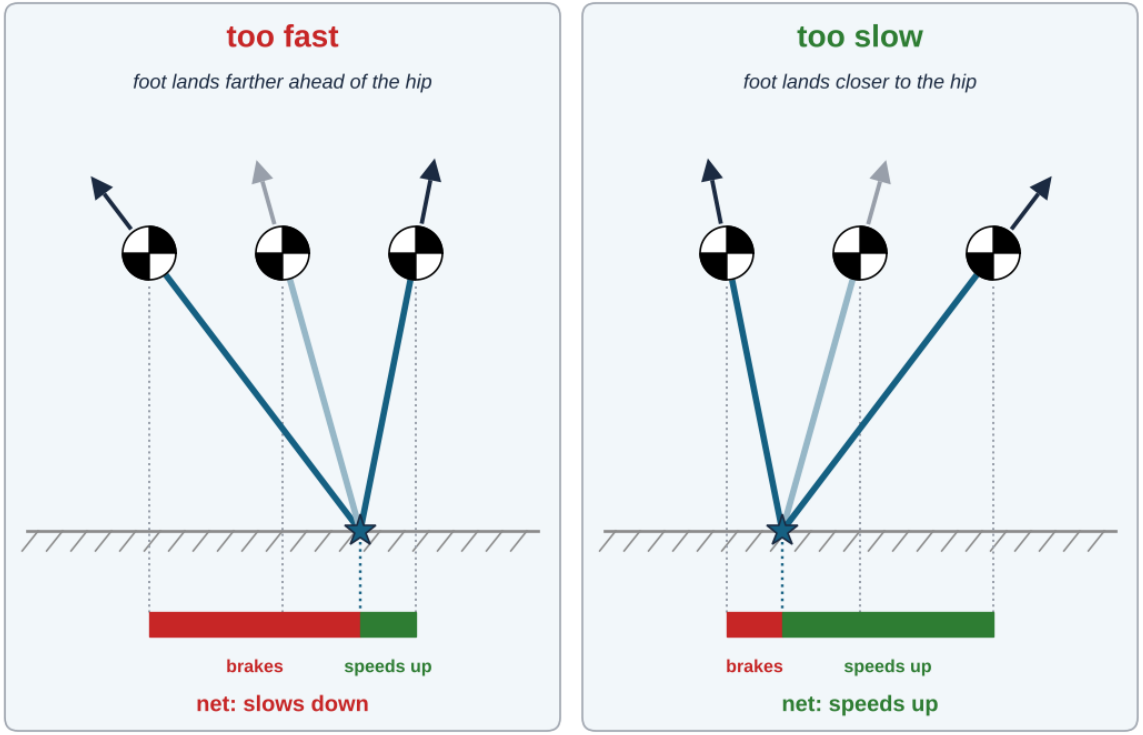
- reference x^{ref}
- MPC plan over the horizon (one dot per step)
- re-solved one step later, horizon shifted
- ground contact force on a stance foot (applied: first step only)

legs drawn: the stance pair only (swing legs are massless and ignored)



Foot placement [2]

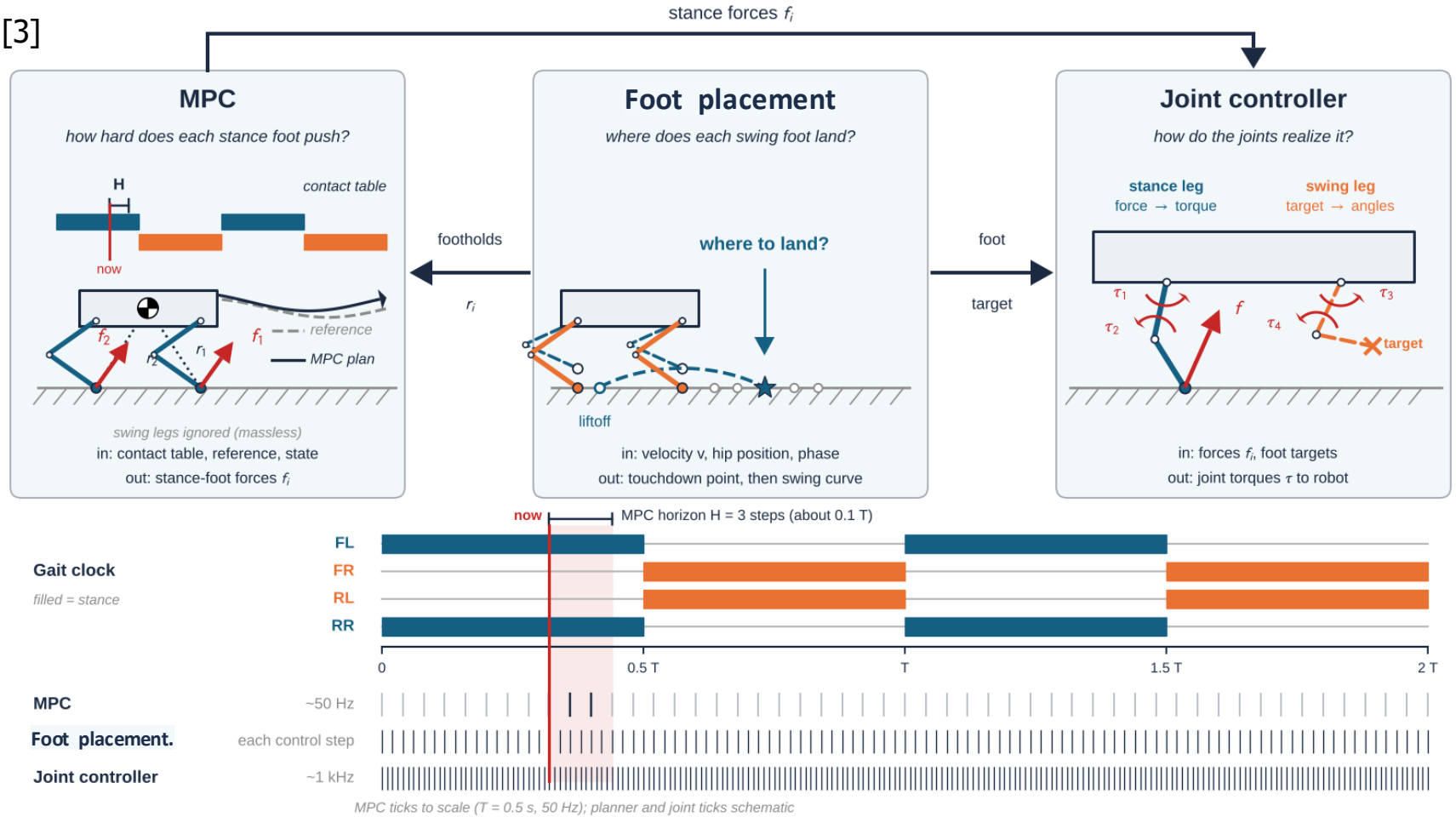




v: measured velocity

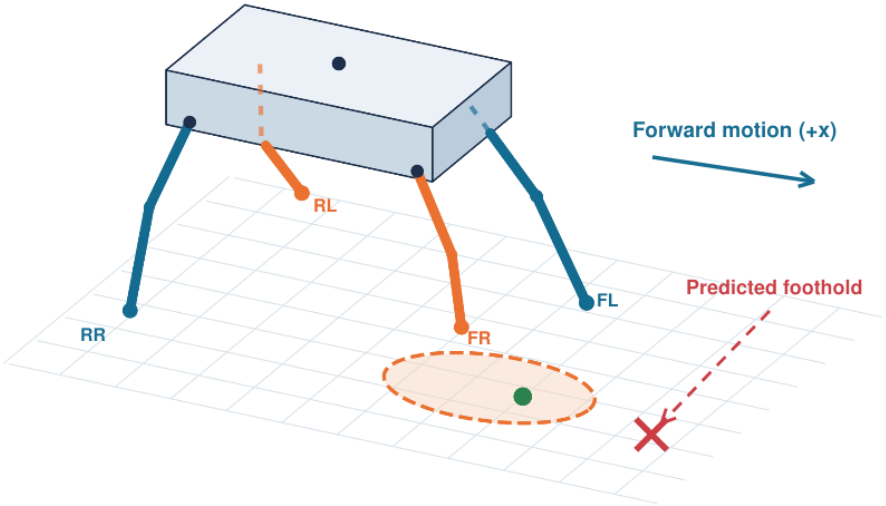
$$p = \frac{vT}{2} + K_p(v - v^{ref})$$

[3]


stance forces f_i
MPC
how hard does each stance foot push?
H
contact table
now
f_2
r_2
r_1
f_1
reference
MPC plan
swing legs ignored (massless)
in: contact table, reference, state
out: stance-foot forces f_i
footholds
r_i
Foot placement
where does each swing foot land?
where to land?
liftoff
in: velocity v, hip position, phase
out: touchdown point, then swing curve
foot
target
Joint controller
how do the joints realize it?
stance leg
force → torque
swing leg
target → angles
τ_1
τ_2
f
τ_4
τ_3
target
in: forces f_i, foot targets
out: joint torques τ to robot
now
MPC horizon H = 3 steps (about 0.1 T)
Gait clock
filled = stance
FL
FR
RL
RR
0
0.5 T
T
1.5 T
2 T
MPC
~50 Hz
Foot placement.
each control step
Joint controller
~1 kHz
MPC ticks to scale (T = 0.5 s, 50 Hz); planner and joint ticks schematic

MOTIVATION / FAILURE MODE 1

Unreachable swing-foot touchdown

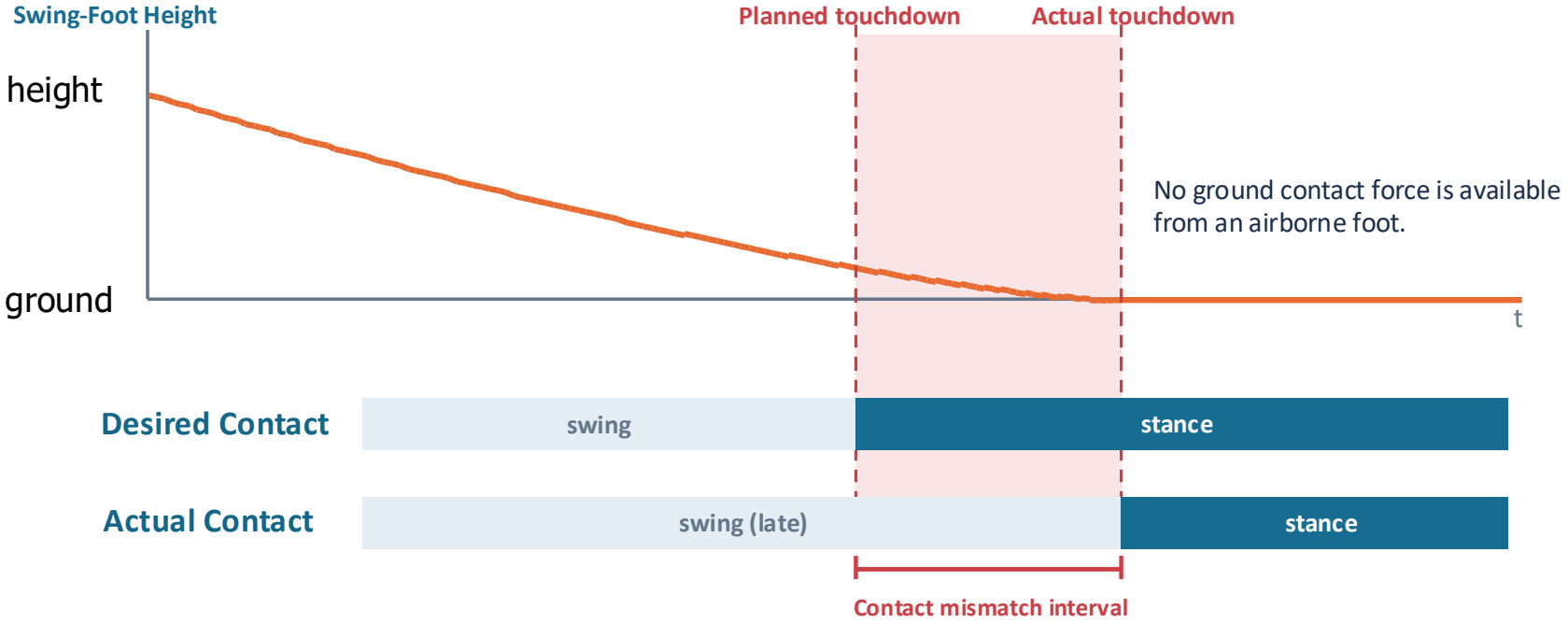


- Orange: swing legs and candidate region
- Red: requested touchdown point

MOTIVATION / FAILURE MODE 2

Touchdown delay creates a contact mismatch

If a swing foot fails to reach the ground at its scheduled touchdown time, the MPC contact assumptions may no longer match the robot's actual support condition.



Periodic Stability Analysis

- Trot is periodic, but the **controller is clocked**: the gait phase ϕ decides which feet are in stance.
- So the closed loop is a **time-varying** system with period T.
- Add the phase as a state $\Rightarrow$ an autonomous system with a periodic orbit.

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \phi), \qquad \dot{\phi} = 1/T, \quad \phi \in [0, 1)$$
$$\boldsymbol{\xi} = (\boldsymbol{x}, \phi) \quad \Rightarrow \quad \dot{\boldsymbol{\xi}} = F(\boldsymbol{\xi}) \quad \text{(autonomous)}$$

- Section: $\Sigma = \{\boldsymbol{\xi} : \phi = 0\}$ (start of each stride).
- Record the state only on Σ: one point per stride.
- Periodic gait $\Rightarrow$ a **fixed point** of the return map.

$$\boldsymbol{s}_{k+1} = P(\boldsymbol{s}_k), \qquad \boldsymbol{s}_k = \boldsymbol{x}(kT) \ \text{(state sampled at } \phi = 0)$$
$$\text{periodic gait:} \quad \boldsymbol{s}^* = P(\boldsymbol{s}^*)$$

- The robot moves forward forever, so x (and y) are not periodic $\Rightarrow$ Reduced coordinates $\boldsymbol{s}$.
- Keep what must come back: height, attitude, body velocities, joint states.

$$\boldsymbol{e}_k = \boldsymbol{s}_k - \boldsymbol{s}^*, \qquad \boldsymbol{e}_{k+1} = P(\boldsymbol{s}^* + \boldsymbol{e}_k) - \boldsymbol{s}^* \approx J\,\boldsymbol{e}_k, \qquad J = \left.\frac{\partial P}{\partial \boldsymbol{s}}\right|_{\boldsymbol{s}^*}$$

Other Candidate Metrics

- **Trajectory tracking error**
- **Region of Attraction**

Disclosure of GenAI Usage:

Images redrawn from hand-drawn sketches via GenAI

Reference

[1] Jared Di Carlo, Patrick M Wensing, Benjamin Katz, Gerardo Bledt, and Sangbae Kim. Dynamic locomotion in the mit cheetah 3 through convex model-predictive control. In 2018 IEEE/RSJ international conference on intelligent robots and systems (IROS), pages 1–9. IEEE, 2018

[2] Marc H Raibert, H Benjamin Brown Jr, and Michael Chepponis. Experiments in balance with a 3d one-legged hopping machine. The International Journal of Robotics Research, 3(2):75–92, 1984.

[3] Joseph Norby, Yanhao Yang, Ardalan Tajbakhsh, Jiming Ren, Justin K Yim, Alexandra Stutt, Qishun Yu, Nikolai Flowers, and Aaron M Johnson. Quad-sdk: Full stack software framework for agile quadrupedal locomotion. In ICRA Workshop on Legged Robots, pages 1–5, 2022.

[4] https://www.animatornotebook.com/learn/quadrupeds-gaits